

Reading the Sensitivity Peak of a PI Loop from a Single Step Response

Abstract

The sensitivity peak M_s tells an engineer how robust a loop is. Measuring it normally needs a plant model or a frequency sweep, and most installed loops have neither. This paper shows that a single ordinary setpoint step already contains the answer, and gives a simple law for reading it.

- *Why is M_s in the record?* For a unit setpoint step the error transform is $E(s) = S(s)/s$. The difference of the record is therefore the impulse response of the sensitivity function itself. No second experiment is needed.
- *How is it read?* Count how many times the error curls around its settling point. Call the count N . We derive the law $M_s = N/(c\sqrt{n_m^2 + n^2})$, where $c = \ln(1/\varepsilon)/2\pi$ is fixed by the counting rule and n, n_m are the loop's phase and magnitude slopes where the phase passes -180° . The distance to instability cancels out of the derivation, which is what makes the reading possible without a model.
- *Why does dead time not disturb it?* Because a dead time contributes the same amount to phase and to phase slope. When the integral time cancels the dominant lag, the crossover condition pins the dead time's phase share, and therefore its slope share, at exactly $\pi/2$ —whatever the dead time is. We compute $n = \pi/2$ for relative dead times from 0.1 to 0.9, to three digits.
- *Why does it fail for PID?* Derivative action supplies phase lead, so the dead time no longer has to supply the whole $-\pi/2$. Its share, and hence n , then depends on how much dead time there is. The invariance is lost by construction, not by accident.

Measurement confirms each prediction with no fitted constant. The dimensionless ratio the law predicts to be unity is measured at 1.256, 1.131 and 1.074 at $M_s \approx 3.5$, 6.5 and 8.5—converging as the theory requires, with interquartile ranges a percent or two wide, and agreeing across plant orders one to three. The map moves by at most 0.064 across relative dead times 0.3 to 0.9 for PI, against 2.58 for PID at $\tau = 0.3$; and readings on fixed classical tunings have median error +5%, with 22 of 24 within $\pm 25\%$. A validity condition read from the same record—does the error change sign—rejects every reading that would have been wrong.

1 Introduction

The sensitivity peak $M_s = \max_\omega |S(j\omega)|$, with $S = 1/(1+CG)$, is the standard measure of how robust a control loop is. A value near 1.4 is comfortable, a value near 2.0 is marginal, and the design window $M_s \in [1.2, 2.0]$ is long established [3, 4].

Getting the number is the problem. It is defined in the frequency domain, so it needs the loop transfer function: either a model, which someone must identify, or a frequency sweep, which someone must commission. Of the eleven thousand loops surveyed by Desborough and Miller [5], almost none have either, and the survey's main finding is that many of them run badly.

What those loops do produce is setpoint steps. This paper shows that one such step is enough. The argument is short:

- The step record *is* the sensitivity function, because $E(s) = S(s)/s$.
- Near the point where a loop becomes marginal, one pole pair dominates. Its damping sets the height of the peak.
- Counting how many times the error curls around its settling point measures that damping.

Putting these together gives a linear law relating the count to M_s (Section 2). The law has one free constant, the loop's phase slope at crossover. Section 3 shows why that constant is fixed at $\pi/2$ for a PI loop no matter how much dead time the plant has, which is the reason the method needs no plant knowledge. Section 4 shows why the same argument fails once derivative action is added. Sections 5 and 6 give the validity condition and the measurements.

We restrict attention to PI control. This is where the result is clean, and it is also where the loops are: derivative action is switched off in most industrial installations [2].

The counting device is the turn index of [1], used there as a damping diagnostic with limits fixed by calibration. Giving it a frequency-domain meaning turns it into a measurement and, as a by-product, replaces that calibration by a specification (Section 7).

2 The law

2.1 The record is the sensitivity function

For a unit setpoint step,

$$E(s) = \frac{S(s)}{s}. \quad (1)$$

The recorded error carries all of S , and its first difference is the impulse response of S . This is why one ordinary step suffices and no dedicated experiment is required.

2.2 Counting the curls

Plot the error against its own difference and watch the trajectory spiral into the point where the loop settles. Let N be the number of turns it makes, counted after normalizing each axis and stopping when the curve enters a small disc of radius ε about that point [1, Def. 1].

A loop that is well damped barely completes a quarter turn. A loop close to instability winds many times. For a mode $e^{-\alpha t} \cos \omega t$ the spiral is logarithmic, and the count is

$$N = c \frac{\omega}{\alpha}, \quad c = \frac{\ln(1/\varepsilon)}{2\pi}, \quad (2)$$

which is [1, Prop. 3]. With the default $\varepsilon = 0.1$, $c = 0.3665$. The count is a pure number: it does not change if the step is made larger or the clock runs faster.

2.3 From count to peak

Near the stability boundary the loop loses stability through a single complex pole pair [1, Prop. 2], at the frequency ω_{180} where the phase passes $-\pi$. Let $m = 1 - |L(j\omega_{180})|$ say how close the loop is to that boundary, and let

$$n = -\frac{d \arg L}{d \ln \omega}, \quad n_m = -\frac{d \ln |L|}{d \ln \omega} \quad (3)$$

be the phase and magnitude slopes there, both dimensionless. Writing $u = \ln(\omega/\omega_{180})$ and expanding in logarithmic coordinates,

$$L(j\omega) \approx -(1-m)e^{-n_m u} e^{-jnu}, \quad (4)$$

so that to first order $1+L \approx m+n_m u+jnu$. Minimising over u gives the distance from the locus to the critical point, and hence

$$M_s = \frac{\sqrt{n_m^2 + n^2}}{mn}. \quad (5)$$

The same expansion locates the pole. Solving $1+L = 0$ near $j\omega_{180}$ gives a pair with frequency $\omega \approx \omega_{180}$ and decay rate $\alpha = m\omega_{180}n/(n^2 + n_m^2)$, so

$$\frac{\omega}{\alpha} = \frac{n^2 + n_m^2}{mn}. \quad (6)$$

Dividing (6) by (5) eliminates m —the distance to the boundary, which the record cannot report—and leaves, with (2),

$$M_s = \frac{\omega/\alpha}{\sqrt{n_m^2 + n^2}} = \frac{N}{c\sqrt{n_m^2 + n^2}} \quad (7)$$

Three things are worth stating plainly. The relation is *linear* in the count. The only plant-dependent quantity in it is the slope pair at one frequency: plant order, lag structure and dead time enter nowhere else. And m has cancelled, so no knowledge of the gain margin is needed—which is what makes the reading possible from a record alone.

Equation (7) is a first-order result in m , hence exact in the limit of a marginal loop. For a comfortable loop $|S|$ does not fall below one far from resonance, and the baseline adds a correction that decays as M_s grows. Section 6 measures how fast.

3 Why dead time does not matter

Equation (7) would be of little use if n varied from plant to plant, since n is not measurable from the step record. For PI control it does not vary, and the reason is elementary.

A dead time L_d contributes phase $-L_d\omega$ and phase slope $-d(-L_d\omega)/d \ln \omega = L_d\omega$. *Its slope contribution equals its phase contribution.* A first-order lag does not share this property: it contributes phase $-\arctan(T\omega)$, approaching $\pi/2$, while its slope $T\omega/(1+T^2\omega^2)$ falls back to zero.

Now take a PI controller whose integral time cancels the dominant lag, $T_i = T$, which is the IMC and SIMC choice [4]. The loop phase is then

$$\arg L(j\omega) = -\frac{\pi}{2} - L_d\omega, \quad (8)$$

the $-\pi/2$ coming from the integrator. The crossover condition $\arg L = -\pi$ therefore forces

$$L_d\omega_{180} = \frac{\pi}{2}, \quad (9)$$

and since the dead time's slope equals its phase,

$$n = L_d\omega_{180} = \frac{\pi}{2} \quad \text{exactly, for every } L_d. \quad (10)$$

A longer dead time simply moves the crossover to a lower frequency. The product $L_d\omega_{180}$, which is what n depends on, does not move. This is the whole reason the reading needs no plant knowledge.

Table 1 confirms (10) numerically, and gives n_m , which sits at unity because the cancelled lag leaves the integrator's single -1 slope.

Table 1: Phase and magnitude slope at crossover, PI with $T_i = T$, and the slope of the law (7) they predict. $\tau = L_d/(L_d + T)$.

τ	0.1	0.3	0.5	0.7	0.9
n	1.57	1.57	1.57	1.57	1.57
n_m	1.00	1.00	1.00	1.00	1.00
ω_{180}	14.14	3.67	1.57	0.67	0.18
predicted slope	1.47	1.47	1.47	1.47	1.47

Remark 1 (Scope). *Equation (10) rests on the cancellation $T_i = T$. Away from it the arctangents no longer cancel and n drifts: over $T_i/T \in [0.5, 8]$ and $\tau \in [0.1, 0.9]$ it ranges from 1.14 to 2.62, so the slope of (7) varies by a factor of about two in the worst case. The law itself is unaffected—only its constant is. Section 6 measures what this costs in practice, which is less than the range suggests, because tunings in use cluster near cancellation.*

4 Why derivative action breaks it

The argument of Section 3 used one fact: the integrator supplies $-\pi/2$ of phase, the lag is cancelled, and so the dead time must supply the remaining $-\pi/2$. Its slope share is then pinned because slope and phase coincide for a dead time.

Derivative action breaks the first step. It supplies phase *lead*, so the dead time no longer has to supply $-\pi/2$; it supplies less, and how much less depends on where the crossover falls, which depends on how much dead time there is. The pinning argument therefore fails, and n becomes a function of τ :

τ	0.1	0.3	0.5	0.7	0.9
n , PI ($T_d = 0$)	1.57	1.57	1.57	1.57	1.57
n , PID ($T_d/T_i = 0.25$)	2.82	2.28	1.47	1.48	1.57

Across $\tau \in [0.3, 0.9]$ the PID phase slope spreads by 48%, against 0% for PI. The failure is not a numerical accident: it follows directly from removing the condition that pinned n .

The theory also says where PID should still work. At large τ the plant's dead time dominates the derivative's lead, and the spread falls to 6% over $\tau \in [0.5, 0.9]$. Measurement agrees: recalibrating the map with $T_d/T_i = 0.25$ shifts it by 0.254 at $\tau = 0.9$ and 0.314 at $\tau = 0.7$ —tolerable—but by 2.583 at $\tau = 0.3$. A PID reading is therefore usable on delay-dominant loops and not on balanced ones, exactly as (10) predicts.

Strong derivative action fails for a second reason: the controller zeros capture the critical pole pair, so the single-dominant-pair picture behind (4) no longer holds. This is the mechanism already noted in [1, Rem. 3], and it is why the map degrades sharply past $T_d/T_i = 0.25$.

5 When the reading is valid

The expansion (4) assumes the dominant pair is visible in the record. Sometimes it is not. If integral action is very slow, the record is dominated by a slow, non-oscillating decay, and the resonant mode rides on it as ripple that never crosses zero. The trajectory then stays on one side of the settling point and cannot wind, whatever the true M_s is.

Because the count is essentially half the number of times the error crosses zero, the condition for a valid reading is as simple as it could be:

The reading is valid when the error record changes sign.

No threshold, no parameter, no model. When the record is one-signed the instrument returns nothing rather than something wrong.

Two facts make this workable. Probing with a load step instead does not help—under the same conditions it gives $N = (0.13, 0.00, 0.00)$, less informative still. And over 497 runs on four process-typical plants, one-signed records never produced a count anywhere near the limits of [1]: their maximum was 0.32 of the limit while true M_s ranged from 1.2 to 6.0. The reading is one-sided. It can say the peak is at least so high; it cannot say the peak is low.

6 Measurements

Every number here tests a prediction made above. Nothing is fitted.

The identity, with no adjustable constant. Equation (7) says the dimensionless ratio

$$R = \frac{M_s c \sqrt{n_m^2 + n^2}}{N} \quad (11)$$

equals one for a marginal loop, and approaches one as M_s grows. Sweeping lag chains of order one to four with dead time, integral times $T_i/T \in \{0.5, 1, 2\}$ and the gain swept:

M_s	1–2	2–3	3–4	6–7	8–
R (median)	1.754	1.562	1.256	1.131	1.074
interquartile	1.74–1.78	1.32–1.75	1.24–1.27	1.12–1.13	1.07–1.08

R falls monotonically towards one, reaching 1.074 at $M_s \approx 8.5$ and still decreasing. This is the behaviour (7) predicts: the identity is exact for a marginal loop, and the excess at moderate M_s is the baseline correction noted above. The interquartile ranges are a percent or two wide, so the agreement is not an average over scatter.

Only the slopes matter. Equation (7) claims the plant enters through n and n_m alone. Breaking the same sweep out by plant order tests it directly:

M_s range	order 1	order 2	order 3
2–3	1.549	1.502	1.384
3–5	1.185	1.189	1.284
5–8	—	1.106	1.154

At each level the three orders agree to within a few percent, and the residual spread is smaller than the drift with M_s . Pooling all orders at $M_s \geq 5$ gives $R = 1.104$ over a range 1.036 to 1.176.

Dead time is invisible, as (10) says. Recalibrating the map at each τ separately and comparing over $N \in [0.3, 1.5]$, the largest shift against the $\tau = 0.5$ reference is 0.041 at $\tau = 0.3$, 0.041 at $\tau = 0.7$ and 0.064 at $\tau = 0.9$. One calibration serves the range, which is (10) restated as a measurement.

Readings on fixed tunings. Four process-typical plants [1], PI controllers with $T_i/L_d \in \{1.5, 2, 3, 4, 6\}$ and four gain levels, one setpoint step each, compared against exact M_s . On valid records: $n = 24$, median error +5%, interquartile range −7% to +8%, and 22 of 24 within $\pm 25\%$. This is the practical cost of Remark 1—the working loop is not marginal, and its T_i does not exactly cancel—and it is modest.

The validity condition catches the failures. In a matched test over five classical tuning rules, every gross error —51%, −47%, −37%, −31%—fell on a record the condition rejects, while accepted readings stayed within −10% to +14%.

7 An application: specifying a tuning rule

The SPIN rule [1] lowers the gain of the lowest band whose count exceeds a limit, with limits $\bar{N} = (0.5, 0.75, 1.0)$ fixed by calibration. Reading them through the law gives peaks of

$$(1.71, 1.92, 2.29), \quad (12)$$

a graded ladder: two inside the classical design window and the third just above it. The calibration landed there without being aimed there.

The limits may therefore be *stated* instead of chosen: pick the admissible peak, invert the law, obtain $\bar{N}$. The specification is also finer than usual, since it is per band rather than one global number.

8 Conclusion

A setpoint step already contains the sensitivity function, because $E(s) = S(s)/s$. Counting how many times the error curls around its settling point measures the damping of the dominant pole pair, and the peak follows by a linear law whose only plant-dependent constant is the phase slope at crossover.

For a PI loop with the lag cancelled, that constant is $\pi/2$ exactly, whatever the dead time, because a dead time contributes equally to phase and to phase slope and the crossover condition pins its share. This is the reason the reading needs no model. Adding derivative action removes the pinning, and with it the invariance—which is why this paper is about PI loops.

Two limitations bound the result. The analysis is noise-free, and noise enters the differenced coordinates directly. And the reading is one-sided: it is valid when the error changes sign and silent otherwise. Both are stated conditions rather than hidden ones, and the second is decided from the same record it qualifies.

Data availability

Scripts reproducing every number are included with the SPIN reference implementation (<https://github.com/ottapav/roboPID-simulator>).

References

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